

Activity 3: Debug and Fix the EMI Calculator using AI

Purpose

Learn how to use AI to identify, analyze, and resolve errors in an existing application. AI can assist developers in understanding error messages, identifying root causes, and recommending fixes.

Pre-requisites

- Completion of Activities 1 and 2
 - Working EMI Calculator application
 - VS Code with Cline configured
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Estimated Duration

40 Minutes

Scenario

The QA team has tested the EMI Calculator and reported several issues. As a developer, your task is to use AI to investigate the problems, identify the root causes, and implement appropriate fixes.

Tasks

Task 1: Analyze the Issue

Run the application and reproduce the issue before making any code changes.

Suggested Prompt

Analyze the current EMI Calculator application before making any changes.

Review the HTML and KnockoutJS ViewModel to identify possible issues that could affect functionality, user experience, or maintainability.

For each issue:

- Describe the problem.
- Explain the possible root cause.
- Recommend an appropriate solution.

Do not modify the code yet.

Task 2: Fix the Identified Issues

After reviewing the analysis, implement the required fixes.

Suggested Prompt

Based on the identified issues, update the Oracle JET application.

Requirements:

- Modify only the required files.
 - Preserve the existing functionality.
 - Explain each change before generating the updated code.
 - Follow Oracle JET and KnockoutJS best practices.
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Task 3: Interpret an Error Message

If the application displays an error during execution, use AI to understand it.

Suggested Prompt

Explain the following error message.

Describe:

- What the error means.
- The probable root cause.
- Which file and code section should be investigated.
- The recommended fix.
- How to prevent similar issues in future.

Explain the answer in simple language suitable for a beginner developer.

Note: Copy and paste the actual error message from the terminal or browser console before running the prompt.

Task 4: Validate the Fix

After implementing the fixes, review the application again.

Suggested Prompt

Review the updated EMI Calculator.

Verify whether the identified issues have been resolved.

Check for:

- Correct functionality
- Code quality
- Side effects introduced by the changes

Recommend any additional improvements if required.

Challenge Task

Improve the application by adding additional input validation.